Core Supervised Learning — Preprocessing, Models & Evaluation

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1. Introduction

This report presents the results of supervised learning experiments on the UCI Adult (Census Income) dataset. The goal is to predict whether an individual earns more than $50,000 per year — a binary classification problem. This is a continuation of Day 1, where baselines were established. In this phase, we implement proper preprocessing pipelines, train two supervised models (Logistic Regression and Decision Tree), and evaluate them using multiple metrics.

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2. Preprocessing Choices

2.1 Feature Identification

- **Numeric features:** `age`, `fnlwgt`, `education-num`, `capital-gain`, `capital-loss`, `hours-per-week`
- **Categorical features:** `workclass`, `education`, `marital-status`, `occupation`, `relationship`, `race`, `sex`, `native-country`

2.2 Missing Value Handling

- **Numeric:** `SimpleImputer(strategy='median')` — chosen because median is robust to outliers.
- **Categorical:** `SimpleImputer(strategy='most_frequent')` — chosen because mode is the safest default for categorical data.

Alternatives considered:

- Mean imputation (rejected due to outliers in `capital-gain`).
- Dropping rows (rejected to preserve data size).
- Constant "Missing" category (rejected because `most_frequent` is simpler and effective).

2.3 Scaling & Encoding

- **Numeric:** `StandardScaler` — required for Logistic Regression to ensure coefficients are comparable.
- **Categorical:** `OneHotEncoder(handle_unknown='ignore')` — chosen because it preserves nominal relationships without assuming order.

Alternatives considered:

- `OrdinalEncoder` (rejected because it would impose false ordering).
- Target encoding (rejected for now — will be tested in Day 3).

2.4 Pipeline Integrity

All preprocessing steps were encapsulated in a `ColumnTransformer` and `Pipeline` to **prevent data leakage**. The preprocessing was fitted **only on the training set** and applied to the hold-out test set.

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3. Model Training

Two models were trained using the preprocessing pipeline:

Model	Key Parameters
Logistic Regression	`random_state=42`, `solver='liblinear'`, `max_iter=1000`
Decision Tree	`random_state=42`, `max_depth=10`

Both models were trained on the **training set only**. The hold-out test set was kept untouched until final evaluation.

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4. Results & Comparison

Model	Accuracy	Precision	Recall	F1	ROC AUC	PR AUC
Logistic Regression	85.28%	74.32%	58.81%	65.66%	**90.40%**	76.27%
Decision Tree	**85.79%**	**79.28%**	55.00%	64.95%	89.95%	**77.14%**
Baseline (Day 1)	75.30%	48.44%	49.70%	49.06%	66.53%	~44%

Key Observations:

- Both models outperform the Day 1 baseline by **10%+ in accuracy** and **~20% in ROC AUC**.
- Logistic Regression has slightly better **ROC AUC (90.40%)**, indicating better overall ranking ability.
- Decision Tree has better **Precision (79.28%)**, meaning it makes fewer false positive errors.

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5. Error Analysis

Model	False Positives	False Negatives
Logistic Regression	475	963
Decision Tree	336	1052

Observation:
Both models have **more false negatives than false positives** — meaning they miss high-income individuals more often than they falsely predict them. For a marketing campaign, **false positives** (wasted budget) are more costly. Decision Tree's lower FP count makes it slightly better for this business context.

6. Interpretability

Logistic Regression — Top Positive Coefficients:

- `capital-gain` (2.23)
- `marital-status_Married-civ-spouse` (1.57)
- `Exec-managerial` (0.79)
- `education-num` (0.76)

Interpretation: Higher capital gain, being married, working in managerial roles, and having more education are strong predictors of high income.

Logistic Regression — Top Negative Coefficients:

- `Priv-house-serv` (-1.41)
- `Never-married` (-1.22)
- `Female` (-1.00)
- `Farming-fishing` (-0.90)

Interpretation: These features are associated with lower income.

Decision Tree — Overfitting Check:

- **Tree Depth:** 10
- **Train Score:** 86.84%
- **Test Score:** 85.79%
- **Overfitting Gap:** 1.05% (minimal)

Top 3 Splits:

1. `Married-civ-spouse`
2. `capital-gain`
3. `education-num`

The tree's logic aligns with domain knowledge and shows no significant overfitting.

7. Model Selection for Day 3

Primary Model: Logistic Regression

Reasons:

- **Interpretability:** Coefficients are easy to explain to stakeholders.
- **Performance:** ROC AUC of 90.40% is strong and close to Decision Tree.
- **Generalization:** No overfitting — train-test gap is minimal.
- **Business Alignment:** Precision (74.32%) is acceptable for a marketing use case.

Secondary Model (Backup): Decision Tree

Reasons:

- Captures non-linear relationships.
- Higher precision (79.28%) and fewer false positives.
- Can be improved with pruning and hyperparameter tuning.

Planned Improvements for Day 3:

- Feature engineering (age bins, interaction terms).
- Outlier handling (capital-gain, capital-loss).
- Target encoding for high-cardinality categoricals.
- Hyperparameter tuning using GridSearchCV.

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8. Conclusion

Both models significantly outperform the Day 1 baseline. Logistic Regression is recommended as the primary model due to its interpretability, strong performance, and generalization. Decision Tree will be retained as a backup. The preprocessing pipeline is robust and reusable. Day 3 will focus on feature engineering and hyperparameter tuning to further improve performance.